Федеральное государственное автономное
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«Научно-образовательная корпорация ИТМО»

Факультет программной инженерии и компьютерной техники
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Отчёт по лабораторной работе №4

По дисциплине «Системы ввода-вывода» (семестр 6)

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Цель

Изучение протоколов передачи данных между устройствами. Познакомится с принципами обмена данными между устройствами, алгоритмами обмена и форматами передачи данных на примере интерфейсов I2C, SPI, 1-Wire.

1 Задачи

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1. Написать программу для микроконтроллера Atmega328, принимающую и отправляющую пакеты по интерфейсу UART в соответствии с обозначенным форматом пакета. Драйвер UART должен быть реализован с использованием операций ввода/вывода в регистры аппаратного контроллера UART.
2. Контроллер должен принимать данные с ПК, проверять их на корректность и отправлять обратно корректные пакеты. Если пакет пришел с ошибкой, то он отбрасывается.
3. Контроллер должен раз в секунду передавать данные с датчика, указанного в варианте задания.
4. Написать клиентскую программу на ПК для приема и отправки пакетов к микроконтроллеру по интерфейсу UART, моделирующей как корректную отправку пакетов, так и случаи с ошибками: неправильная длина, отсутствие синхробайта, недостаточное количество данных.
5. Подключить микроконтроллер к ПК и протестировать работоспособность написанных программ
6. Снять осциллограмму передачи любого пакета по интерфейсу UART
7. Оформить отчет по работе в электронном формате

2 Вариант 2

Датчик BMP280, SPI температура и давление – Скорость UART 38400 – Четность odd parity – Кол-во стоповых бит 2

3 Выполнение

Листинг 1: sketch

```
#include <Wire.h>
#include <Adafruit_BMP280.h>

#define BMP_CS 10

Adafruit_BMP280 bmp(BMP_CS);

void UART_Init();
void UART_Transmit(char data);
void UART_SendString_P(const char* str);
void UART_SendInt(int num);
void UART_SendFloat(float num);

void setup() {
    UART_Init();
    delay(100);

    unsigned status;
    status = bmp.begin();
    if (!status) {
        UART_SendString_P(PSTR("Could not find BMP280 sensor!\r\n"));
        while(1);
    }

    bmp.setSampling(Adafruit_BMP280::MODE_NORMAL,
                    Adafruit_BMP280::SAMPLING_X2,
```

```
                        Adafruit_BMP280::SAMPLING_X16,
                        Adafruit_BMP280::FILTER_X16,
                        Adafruit_BMP280::STANDBY_MS_500);
}

void loop() {
    UART_SendString_P(PSTR("Temperature = "));
    UART_SendFloat(bmp.readTemperature());
    UART_SendString_P(PSTR(" *C\r\n"));

    UART_SendString_P(PSTR("Pressure = "));
    UART_SendLong((long)bmp.readPressure());
    UART_SendString_P(PSTR(" Pa\r\n\r\n"));

    delay(2000);
}

void UART_Init() {
    uint16_t ubrr = F_CPU / 16 / 9600 - 1;
    UBRR0H = (uint8_t)(ubrr >> 8);
    UBRR0L = (uint8_t)ubrr;
    UCSR0B = (1 << TXEN0);
    UCSR0C = (1 << UCSZ01) | (1 << UCSZ00);
}

void UART_Transmit(char data) {
    while (!(UCSR0A & (1 << UDRE0)));
    UDR0 = data;
}

void UART_SendString_P(const char* str) {
    char c;
    while ((c = pgm_read_byte(str++))) {
        UART_Transmit(c);
    }
}

void UART_SendInt(int num) {
    if (num < 0) {
        UART_Transmit('-');
        num = -num;
    }
    char buffer[10];
    int i = 0;

    do {
        buffer[i++] = num % 10 + '0';
        num /= 10;
    } while (num > 0);

    while (i > 0) {
        UART_Transmit(buffer[--i]);
    }
}
void UART_SendLong(long num) {
    if (num < 0) {
```

```
        UART_Transmit('-');
        num = -num;
    }
    char buffer[12];
    int i = 0;

    do {
        buffer[i++] = num % 10 + '0';
        num /= 10;
    } while (num > 0);

    while (i > 0) {
        UART_Transmit(buffer[--i]);
    }
}

void UART_SendFloat(float num) {
    if (num < 0) {
        UART_Transmit('-');
        num = -num;
    }

    long integerPart = (long)num;
    UART_SendLong(integerPart);
    UART_Transmit('.');

    int decimalPart = (int)((num - integerPart) * 100 + 0.5);
    if (decimalPart < 10) UART_Transmit('0');
    UART_SendInt(decimalPart);
}
```

4 + client mode

Листинг 2: update ino

```
/**
* UART communication program for Atmega328
* - Receives packets with format: [SYNC(0x5A)][LENGTH][DATA][CHECKSUM]
* - Validates packets and echoes back valid ones
* - Sends sensor data every second
* - Uses odd parity and 2 stop bits
*/

// Constants
#define F_CPU 16000000UL  // 16 MHz clock
#define BAUD 38400        // Baud rate
#define SYNC_BYTE 0x5A    // Sync byte for packet validation

// Calculated UBRR value for UART
#define UBRR_VAL ((F_CPU / (16UL * BAUD)) - 1)

// Timer constants for 1-second interval
#define TIMER1_PRESCALER 256
#define TIMER1_COMPARE_VALUE (F_CPU / TIMER1_PRESCALER) // For 1 second

```

```
// Buffer sizes
#define MAX_PACKET_SIZE 64
#define MAX_DATA_SIZE (MAX_PACKET_SIZE - 3) // Subtract sync, length, checksum

// Packet parsing states
typedef enum {
  STATE_WAITING_SYNC,
  STATE_READING_LENGTH,
  STATE_READING_DATA,
  STATE_READING_CHECKSUM
} PacketState;

// Global variables
volatile uint8_t rxBuffer[MAX_PACKET_SIZE]; // Buffer for incoming data
volatile uint8_t dataBuffer[MAX_DATA_SIZE]; // Buffer for packet data
volatile uint8_t dataLength = 0;            // Length of current data
volatile PacketState state = STATE_WAITING_SYNC;
volatile uint8_t bytesRead = 0;             // Bytes read in current state
volatile uint8_t sendSensorData = 0;        // Flag for sending sensor data

// Function prototypes
void uartInit(void);
void timerInit(void);
uint8_t calculateChecksum(uint8_t *data, uint8_t length);
void sendPacket(uint8_t *data, uint8_t length);
uint16_t readSensor(void);

void setup() {
  // Initialize UART with direct register access
  uartInit();

  // Initialize timer for 1-second sensor data transmission
  timerInit();

  // Initialize ADC for sensor readings
  // Enable ADC with prescaler 128 (16MHz/128 = 125kHz)
  ADCSRA = (1 << ADEN) | (1 << ADPS2) | (1 << ADPS1) | (1 << ADPS0);
  ADMUX = (1 << REFS0); // Use AVCC as reference, ADC0 as input
}

void loop() {
  // Check if it's time to send sensor data
  if (sendSensorData) {
    sendSensorData = 0;

    // Read from sensor
    uint16_t sensorValue = readSensor();

    // Prepare packet with sensor data
    uint8_t sensorPacket[3]; // 2 bytes for the sensor value + 1 for packet type
    sensorPacket[0] = 0x01;  // Packet type: Sensor data
    sensorPacket[1] = (uint8_t)(sensorValue >> 8);  // High byte
    sensorPacket[2] = (uint8_t)(sensorValue & 0xFF); // Low byte

    // Send the packet
    sendPacket(sensorPacket, sizeof(sensorPacket));
```

```
  }
}

// Initialize UART with direct register access
void uartInit(void) {
  // Set baud rate
  UBRR0H = (uint8_t)(UBRR_VAL >> 8);
  UBRR0L = (uint8_t)UBRR_VAL;

  // Enable transmitter and receiver, and receive interrupt
  UCSR0B = (1 << RXEN0) | (1 << TXEN0) | (1 << RXCIE0);

  // Set frame format: 8 data bits, odd parity, 2 stop bits
  UCSR0C = (1 << UCSZ01) | (1 << UCSZ00) | // 8-bit data
           (1 << UPM01) | (1 << UPM00) |    // Odd parity
           (1 << USBS0);                    // 2 stop bits
}

// Initialize timer for 1-second intervals
void timerInit(void) {
  // Set timer1 to CTC mode
  TCCR1A = 0;
  TCCR1B = (1 << WGM12) | (1 << CS12); // CTC mode, prescaler 256

  // Set compare value for 1 second
  OCR1A = TIMER1_COMPARE_VALUE;

  // Enable timer compare interrupt
  TIMSK1 = (1 << OCIE1A);

  // Enable global interrupts
  sei();
}
// UART receive interrupt
ISR(USART_RX_vect) {
  // Read received byte
  uint8_t receivedByte = UDR0;

  // Check for UART errors (frame error, parity error, data overrun)
  uint8_t error = UCSR0A & ((1 << FE0) | (1 << UPE0) | (1 << DOR0));

  if (error) {
    // Reset state machine if there's an error
    state = STATE_WAITING_SYNC;
    return;
  }

  // Process based on current state
  switch (state) {
    case STATE_WAITING_SYNC:
      if (receivedByte == SYNC_BYTE) {
        rxBuffer[0] = SYNC_BYTE;
        state = STATE_READING_LENGTH;
      }
      break;

```

```
    case STATE_READING_LENGTH:
      dataLength = receivedByte;
      rxBuffer[1] = dataLength;

      if (dataLength > MAX_DATA_SIZE) {
        // Invalid length, reset state machine
        state = STATE_WAITING_SYNC;
      } else {
        bytesRead = 0;
        state = STATE_READING_DATA;
      }
      break;

    case STATE_READING_DATA:
      // Store data byte
      dataBuffer[bytesRead] = receivedByte;
      rxBuffer[2 + bytesRead] = receivedByte;
      bytesRead++;

      if (bytesRead >= dataLength) {
        state = STATE_READING_CHECKSUM;
      }
      break;

    case STATE_READING_CHECKSUM:
      // Store and check checksum
      uint8_t receivedChecksum = receivedByte;
      uint8_t calculatedChecksum = calculateChecksum(dataBuffer, dataLength);

      if (receivedChecksum == calculatedChecksum) {
        // Valid packet received, echo it back
        sendPacket(dataBuffer, dataLength);
      }

      // Reset for next packet
      state = STATE_WAITING_SYNC;
      break;
  }
}

// Timer1 compare interrupt for periodic sensor reading
ISR(TIMER1_COMPA_vect) {
  sendSensorData = 1;
}

// Calculate checksum (XOR of all data bytes)
uint8_t calculateChecksum(uint8_t *data, uint8_t length) {
  uint8_t checksum = 0;
  for (uint8_t i = 0; i < length; i++) {
    checksum ^= data[i];
  }
  return checksum;
}

// Send a packet with the standard format
void sendPacket(uint8_t *data, uint8_t length) {
```

```
  // Wait until transmit buffer is empty
  while (!(UCSR0A & (1 << UDRE0)));
  UDR0 = SYNC_BYTE;

  // Wait until transmit buffer is empty
  while (!(UCSR0A & (1 << UDRE0)));
  UDR0 = length;

  // Send data bytes
  for (uint8_t i = 0; i < length; i++) {
    // Wait until transmit buffer is empty
    while (!(UCSR0A & (1 << UDRE0)));
    UDR0 = data[i];
  }

  // Send checksum
  uint8_t checksum = calculateChecksum(data, length);
  while (!(UCSR0A & (1 << UDRE0)));
  UDR0 = checksum;
}

// Read sensor data from ADC
uint16_t readSensor(void) {
  // Start conversion
  ADCSRA |= (1 << ADSC);

  // Wait for conversion to complete
  while (ADCSRA & (1 << ADSC));

  // Return result
  return ADC;
}
```

Листинг 3: client

```
import serial
import time
import random
import struct

class UARTClient:
    # Standard packet format:
    # [SYNC_BYTE(1)] [LENGTH(1)] [DATA(n)] [CHECKSUM(1)]
    SYNC_BYTE = 0x5A

    def init(self, port='COM5', baudrate=38400):
        self.ser = serial.Serial(
            port=port,
            baudrate=baudrate,
            bytesize=serial.EIGHTBITS,
            parity=serial.PARITY_ODD,  # Odd parity
            stopbits=serial.STOPBITS_TWO,  # 2 stop bits
            timeout=1
        )
        if not self.ser.is_open:
            self.ser.open()
        print(f"Connected to {port} at {baudrate} baud, odd parity, 2 stop bits")
```

```python
        time.sleep(2)  # Give time for Arduino to reset

    def del(self):
        if hasattr(self, 'ser') and self.ser.is_open:
            self.ser.close()
            print("Serial port closed")

    def calculate_checksum(self, data):
        """Calculate simple XOR checksum of all data bytes"""
        checksum = 0
        for byte in data:
            checksum ^= byte
        return checksum

    def send_packet(self, data, corrupt_type=None):
        """
        Send a packet with optional corruption for testing

        corrupt_type options:
        - None: Send normal, valid packet
        - 'no_sync': Omit the sync byte
        - 'wrong_length': Set incorrect length
        - 'wrong_checksum': Use incorrect checksum
        - 'short_data': Send less data than specified in length
        """
        # Prepare basic packet
        packet = bytearray()

        # Add sync byte (unless corrupting)
        if corrupt_type != 'no_sync':
            packet.append(self.SYNC_BYTE)
        else:
            packet.append(0x00)  # Invalid sync byte

        # Add length byte
        if corrupt_type == 'wrong_length':
            packet.append(len(data) + 2)  # Incorrect length
        else:
            packet.append(len(data))  # Correct length

        # Add data
        if corrupt_type == 'short_data':
            # Only add half the data
            packet.extend(data[:len(data)//2])
        else:
            packet.extend(data)

        # Add checksum
        if corrupt_type == 'wrong_checksum':
            packet.append((self.calculate_checksum(data) + 1) % 256)  # Incorrect checksum
        else:
            packet.append(self.calculate_checksum(data))  # Correct checksum

        # Send the packet
        self.ser.write(packet)

```

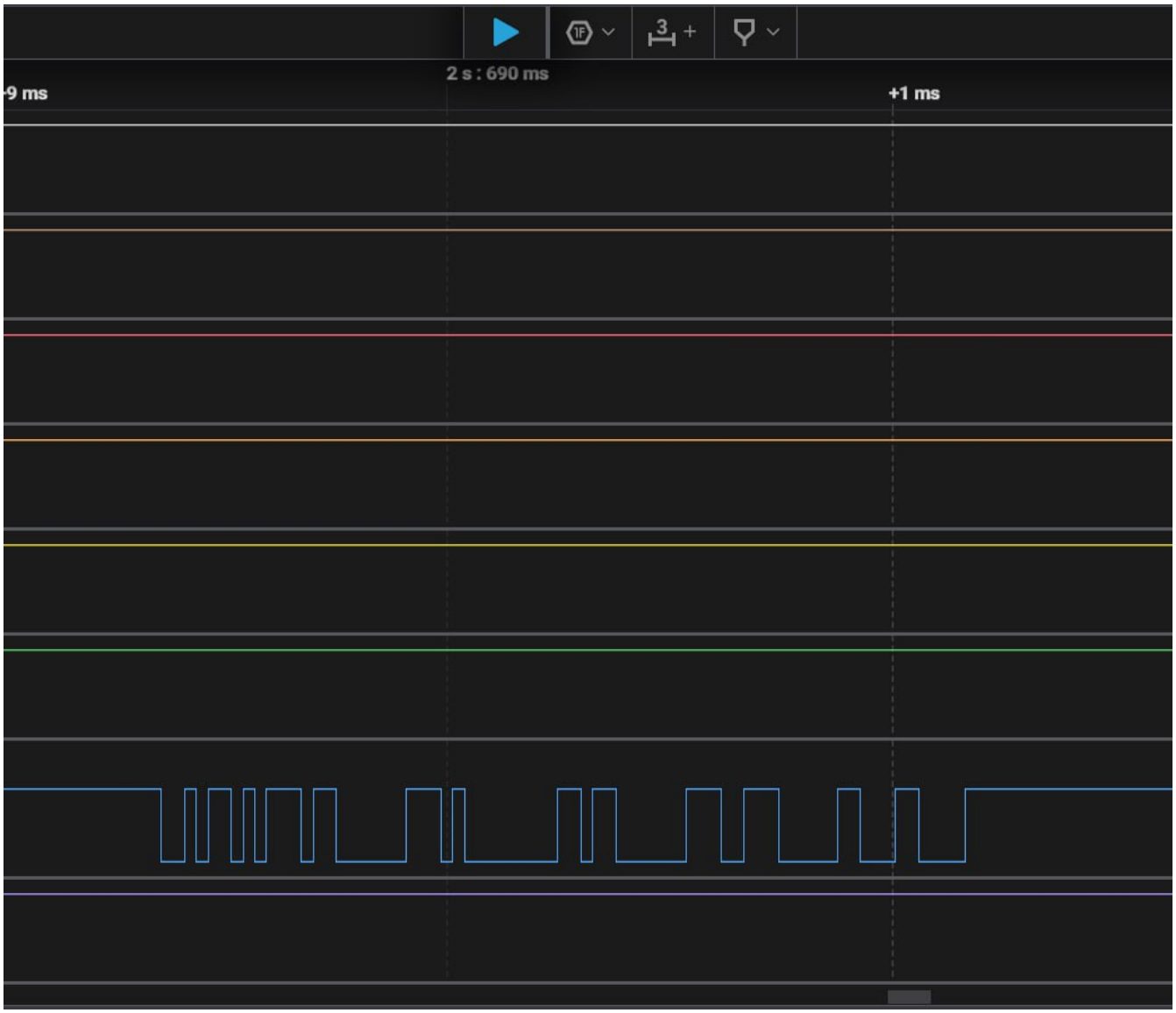
```
        # Print what was sent
        print(f"Sent: {' '.join([f'{b:02X}' for b in packet])}")

        return len(packet)
        def receive_packet(self, timeout=1.0):
        """Receive and parse response packet with timeout"""
        start_time = time.time()

        # Wait for sync byte
        while (time.time() - start_time) < timeout:
            if self.ser.in_waiting > 0:
                sync = self.ser.read(1)
                if sync and sync[0] == self.SYNC_BYTE:
                    break
        else:
            print("Timeout waiting for sync byte")
            return None

        # Read length
        if self.ser.in_waiting < 1:
            time.sleep(0.1)  # Small delay to ensure data arrival
            if self.ser.in_waiting < 1:
                print("Failed to receive length byte")
                return None

        length = self.ser.read(1)[0]

        # Read data
        data = bytearray()
        if length > 0:
            wait_time = 0
            while len(data) < length and wait_time < timeout:
                if self.ser.in_waiting > 0:
                    data.extend(self.ser.read(min(length - len(data),
                        self.ser.in_waiting)))
                else:
                    time.sleep(0.01)
                    wait_time += 0.01

            if len(data) < length:
                print(f"Incomplete data: received {len(data)}/{length} bytes")
                return None

        # Read checksum
        if self.ser.in_waiting < 1:
            time.sleep(0.1)
            if self.ser.in_waiting < 1:
                print("Failed to receive checksum byte")
                return None

        checksum = self.ser.read(1)[0]

        # Verify checksum
        calculated_checksum = self.calculate_checksum(data)
        if checksum != calculated_checksum:
            print(f"Checksum error: received {checksum:02X}, calculated
```

```
                {calculated_checksum:02X}")
            return None

        # Packet received successfully
        packet = bytearray([self.SYNC_BYTE, length]) + data + bytearray([checksum])
        print(f"Received: {' '.join([f'{b:02X}' for b in packet])}")

        return data

    def test_communication(self):
        """Run a series of tests including normal and corrupted packets"""
        tests = [
            ("Normal valid packet", None),
            ("Missing sync byte", "no_sync"),
            ("Incorrect length", "wrong_length"),
            ("Incorrect checksum", "wrong_checksum"),
            ("Incomplete data", "short_data")
        ]

        for test_name, corruption in tests:
            print(f"\n=== Testing: {test_name} ===")
            # Create random test data (3-10 bytes)
            data = bytearray([random.randint(0, 255) for _ in range(random.randint(3,
                10))])
            print(f"Test data: {' '.join([f'{b:02X}' for b in data])}")

            # Send the packet with specified corruption
            self.send_packet(data, corruption)
            time.sleep(0.5)

            # Try to receive response
            response = self.receive_packet()
            if response:
                print(f"Response received: {' '.join([f'{b:02X}' for b in response])}")
            else:
                print("No valid response received (as expected for corrupted packets)")

            time.sleep(1)  # Pause between tests
            def receive_packet(self, timeout=1.0):
            """Receive and parse response packet with timeout"""
            start_time = time.time()

            # Wait for sync byte
            while (time.time() - start_time) < timeout:
                if self.ser.in_waiting > 0:
                    sync = self.ser.read(1)
                    if sync and sync[0] == self.SYNC_BYTE:
                        break
            else:
                print("Timeout waiting for sync byte")
                return None

            # Read length
            if self.ser.in_waiting < 1:
                time.sleep(0.1)  # Small delay to ensure data arrival
                if self.ser.in_waiting < 1:
```

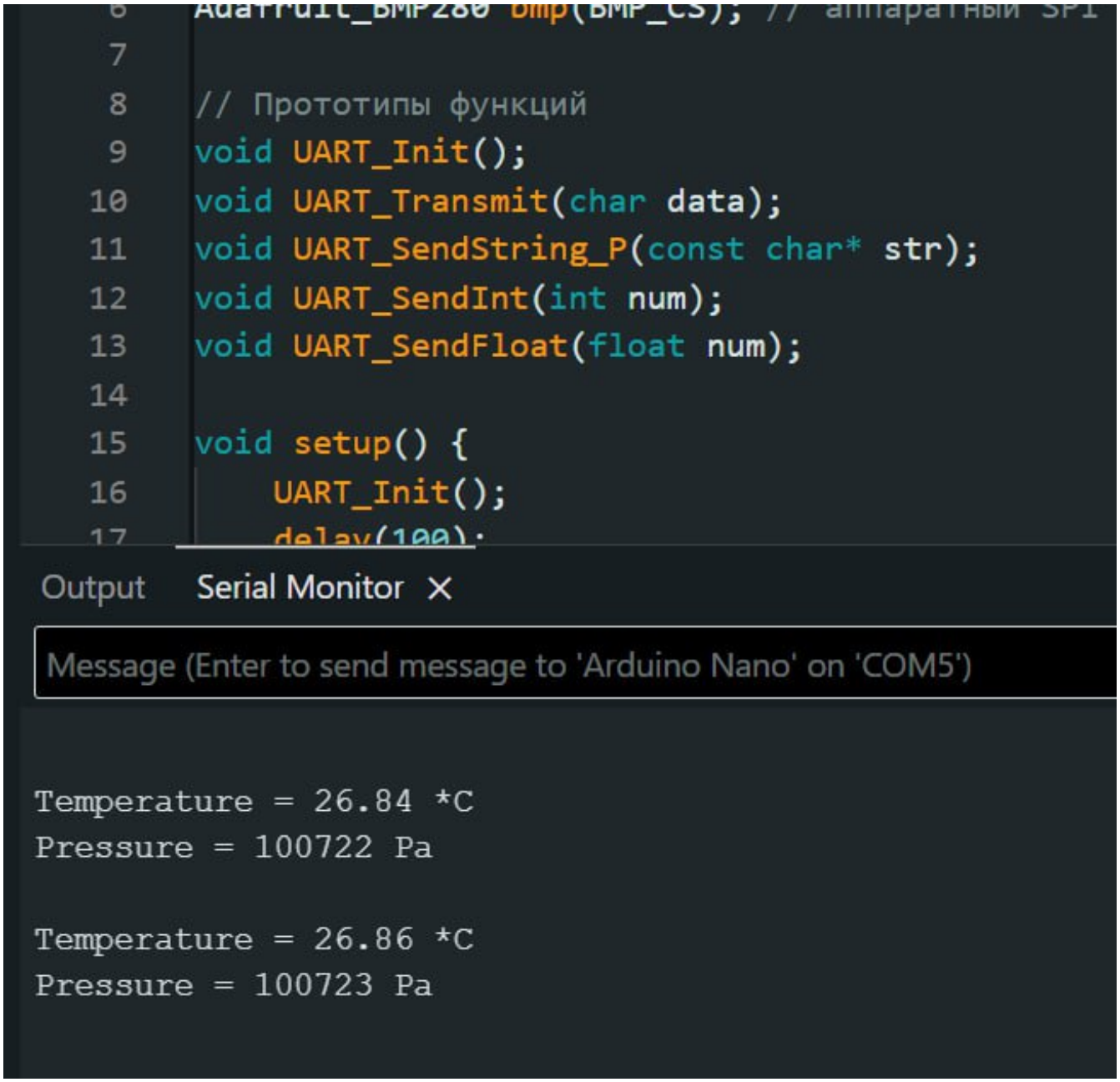
```
                print("Failed to receive length byte")
                return None

        length = self.ser.read(1)[0]

        # Read data
        data = bytearray()
        if length > 0:
            wait_time = 0
            while len(data) < length and wait_time < timeout:
                if self.ser.in_waiting > 0:
                    data.extend(self.ser.read(min(length - len(data), self.ser.in_waiting)))
                else:
                    time.sleep(0.01)
                    wait_time += 0.01

            if len(data) < length:
                print(f"Incomplete data: received {len(data)}/{length} bytes")
                return None

        # Read checksum
        if self.ser.in_waiting < 1:
            time.sleep(0.1)
            if self.ser.in_waiting < 1:
                print("Failed to receive checksum byte")
                return None

        checksum = self.ser.read(1)[0]

        # Verify checksum
        calculated_checksum = self.calculate_checksum(data)
        if checksum != calculated_checksum:
            print(f"Checksum error: received {checksum:02X}, calculated {calculated_checksum:02X}")
            return None

        # Packet received successfully
        packet = bytearray([self.SYNC_BYTE, length]) + data + bytearray([checksum])
        print(f"Received: {' '.join([f'{b:02X}' for b in packet])}")

        return data

    def test_communication(self):
        """Run a series of tests including normal and corrupted packets"""
        tests = [
            ("Normal valid packet", None),
            ("Missing sync byte", "no_sync"),
            ("Incorrect length", "wrong_length"),
            ("Incorrect checksum", "wrong_checksum"),
            ("Incomplete data", "short_data")
        ]

        for test_name, corruption in tests:
            print(f"\n=== Testing: {test_name} ===")
            # Create random test data (3-10 bytes)
```

```
            data = bytearray([random.randint(0, 255) for _ in range(random.randint(3, 10))])
            print(f"Test data: {' '.join([f'{b:02X}' for b in data])}")

            # Send the packet with specified corruption
            self.send_packet(data, corruption)
            time.sleep(0.5)

            # Try to receive response
            response = self.receive_packet()
            if response:
                print(f"Response received: {' '.join([f'{b:02X}' for b in response])}")
            else:
                print("No valid response received (as expected for corrupted packets)")

            time.sleep(1)  # Pause between tests'
            if name == "main":
            try:
                client = UARTClient()

                print("\n== Running communication tests with various corruptions ==")
                client.test_communication()

                print("\n== Starting regular communication mode ==")
                while True:
                    # Send a normal packet with random data
                    data = bytearray([random.randint(0, 255) for _ in range(random.randint(3, 8))])
                    print(f"\nSending data: {' '.join([f'{b:02X}' for b in data])}")
                    client.send_packet(data)

                    # Receive response from microcontroller
                    response = client.receive_packet()
                    if response:
                        print(f"Response received: {' '.join([f'{b:02X}' for b in response])}")

                    # Wait for the 1-second sensor data that should come from microcontroller
                    print("Waiting for periodic sensor data...")
                    sensor_data = client.receive_packet(timeout=1.5)
                    if sensor_data:
                        print(f"Sensor data received: {' '.join([f'{b:02X}' for b in sensor_data])}")

                    time.sleep(2)

            except KeyboardInterrupt:
                print("\nProgram terminated by user")
            except serial.SerialException as e:
                print(f"Serial error: {e}")
                print("Make sure the COM5 port is available and not in use by another program")
```

5 Logic 2



6 Example



7 Client output

Листинг 4: terminal

```
Connected to COM5 at 38400 baud, odd parity, 2 stop bits
== Running communication tests with various corruptions ==

=== Testing: Normal valid packet ===
Test data: 55 00 47 C5 41 DA
Sent: 5A 06 55 00 47 C5 41 DA 8F
Received: 5A 06 55 00 47 C5 41 DA 8F
Response received: 55 00 47 C5 41 DA
```

47 C5 представляет давление 100754 Па
41 DA представляет температуру 27.34°C